

Technical Operations Manual: OSP Engineering & Construction Standards

1. Engineering Deliverables and Project Framework

Standard Deliverable Checklist

Every project submission shall include the following items in the specified format.

Item Name	Format	Quality Expectation
Overview Key Map	PDF	D-Sized (36"x48") providing a holistic view of the project footprint.
Construction Prints	PDF	11"x17" at 1:100 scale; Construction Letter Call Outs must be on the Right side.
Engineering Source	DWG/SHP	Complete CAD files with all layers and LISP routines applied.
Project Workbook	XLSX	Comprehensive 13-tab dataset consistent with Lumos naming conventions.
KMZ Files	KMZ	Geospatial data for the total polygon area AND the Node polygon breakdown.
Permit Package	PDF	Applications including directional bore, missile, or plow verbiage for UG permits.
Plats	PDF/IMG	Supplied for each polygon found during the engineering stage.

The "Workbook" Architecture

The Project Workbook is the financial and logical repository of the design. Designers shall populate the following 13 essential tabs:

1. **Cover Sheet:** Project name, number, date, permits required, and total aerial/buried footage.
2. **Map Tab:** High-level job overview map.
3. **Materials:** Standardized list with highlighted part numbers and quantities.
4. **Labor Sheet:** Input tab for items not auto-populated by the span sheet.
5. **House Count:** Population of all highlighted fields for serviceability.

6. **Cable Sheet:** Sequential list of every cable segment, length, and count.
7. **Labor Span Sheet:** Calculation of labor costs based on conduit type and footage.
8. **MEW (Most Efficient Way):** Critical routing optimization documentation.
9. **Primary Splicing Breakdown:** Granular details for 1x8 hierarchy.
10. **Secondary Splicing Breakdown:** Granular details for 1x4 hierarchy.
11. **Final Address List:** Filtered export of serviceable addresses from the House Count.
12. **Permits & Plats:** Record of approved municipal documents and property boundaries.
13. **Ring Splicing:** Detailed logic for transport and backbone fiber paths.

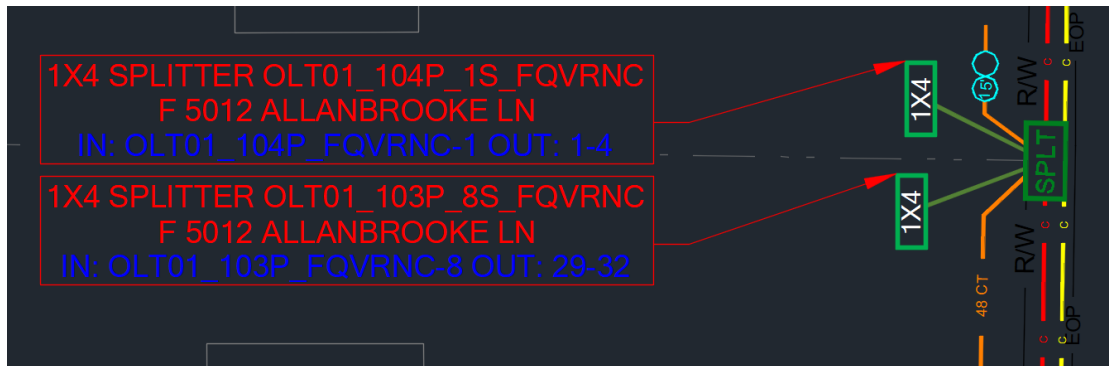
These high-level deliverables provide the structural data necessary for the drafting standards that dictate field safety and permit approval.

Workbook and CX/LLD Steps

The workbook is the financial "source of truth." Designers shall follow the 13-step LLD/CX design roadmap to ensure data integrity:

1. **Count Verification:** Confirm incoming feeder ranges from upstream jobs/cabinets.
2. **Feeder Allocation:** Mapping fibers to geographic sections (top, bottom, southeast, etc.).
3. **Splitter Numbering:** Assigning unique IDs (e.g., 152P) to 1x8 hardware.
4. **Splicing Strategy:** Establishing the "Color for Color" matrix and tube assignments.
5. **Reserve Designation:** Identifying growth fibers and routing them to handholes.

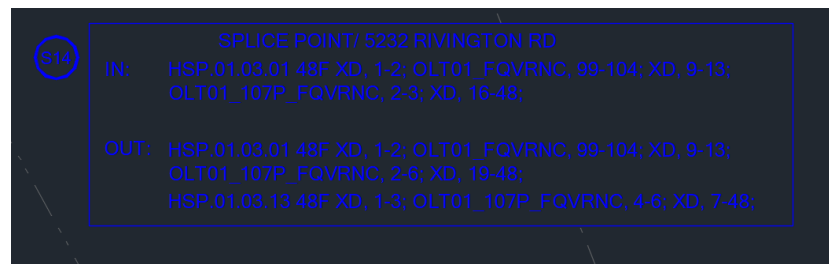
6. **1x4 Hierarchical Assignment:** Numbering secondary splitters (e.g., 152P-1 through 152P-8).



7. **Fiber Callout Documentation:** Creating text callouts for every segment on a continuous line.



8. **Splice Callout Boxes:** Populating blue boxes with IN/OUT cable names and counts.



9. **Cable Sheet Documentation:** Listing every segment, length, and count sequentially.
10. **Labor Span Population:** Entering span footages and conduit types.
11. **Material Sheet Completion:** Fiber is auto-populated; splitters and HHs are manually verified.
12. **House Count Export:** Final serviceability mapping for the IT stack.
13. **Final QC and Package Release:** Final validation before the "Approved for Construction" release.

2. CAD Drafting and Construction Print Standards

Visual clarity and drafting precision directly impact municipal permit approval speed and field safety. Designers shall not deviate from the following standards to ensure all prints are professional, legible, and constructible.

Layer and Annotation Rigor

- **Text Mandate:** All text—including street names, callouts, and addresses—shall be in **ALL CAPS**.
- **Abbreviations:** Street abbreviations shall not include periods (e.g., use "DR" not "DR.").
- **Street Names:** Must be **CENTERED** between Edge of Pavement (EOP) lines and aligned parallel to the EOP.
- **Background Masks:** Mandated for all addresses, footage labels, and sheet titles to ensure legibility.
- **Linetype Synchronization:** The font size for Edge of Pavement (EOP) and Right of Way (ROW) linetypes must be identical.
- **Linework Integrity:** Parcel lines shall be trimmed to the ROW; EOP corners must be filleted; utility lines shall be turned off on construction prints.
- **Pole Attachments:** All poles must show the owner, Pole ID, Lat/Long, and Lumos approval status.

The Visual Hierarchy (V5 Color Standards)

Designers shall utilize the V5 standard color hierarchy. This hierarchy supersedes all prior versions (Ver-9).

Element Category	Color / Symbol	Application Rule
Splice Callouts	Blue	Standard blue box callouts at all cable junctions.
Splitter Callouts	Red	Denotes 1x8 or 1x4 blocks; IN/OUT counts inside must be Blue.
Terminal Boundary	Green	Polygons identifying the 1x4 service area.
Fiber (Underground)	Red	Continuous lines representing the buried FOC path.

Fiber (Aerial)	Blue	Represents strand-mounted fiber.
Fiber (Strand)	Brown	Represents the supporting strand infrastructure.
Conduit	Black Dashed	New conduit routing (1.25" or 2" sizes).
Existing Facilities	Black	Previously installed infrastructure to remain.

Viewport and Scaling Protocol

Consistency is mandatory. Designers shall use the **ALIGNSPACE** command to ensure the scale is identical across all sheets.

- **Matchlines:** Must align perfectly with the viewport border as a continuous line with no redundant data overlap.
- **Centering:** Matchlines shall be **CENTERED on the street** rather than on conduit runs to maintain geographical context.
- **See Sheet/Match Sheet:** All cables leaving a page must have verified pairs.

CAD Best Practices & LISP Integration

Before submission, the following technical steps are mandatory:

- **AutoRotate180:** Must be run on all handholes, footages, and road names to ensure they are oriented right-side up.
- **Auto Dimensions:** The LISP is a starting point; designers shall manually delete incorrect dimensions and ensure all tic marks snap precisely to linework (EOP, ROW, or Conduit).
- **SAVETIME Configuration:** The SAVETIME interval shall be set to **1 minute**.

3. Civil Infrastructure and Design Constraints

Inefficient infrastructure placement leads to massive budget overruns. Every placement must be evaluated for financial and construction logic.

Handhole and Flower Pot Specifications

Hardware must be sized appropriately for its equipment load.

HH Size	Use Case	Maximum Cable Capacity	Hardware/Notes
13x24	SPLT	1 Distro Cable (In/Out)	Fits (1) G5 splice case.
17x30	SM	2 Distro Cables	Fits (1) G5 case; for small splices/MSTs.
24x36	MED	3 Fiber Cables	Fits Tyco 450D; 144 and 288 LCP placement.
30x48	LHH	4 Fiber Cables	432 LCP placement only or feeder ring routes.

Conduit Placement and Road Bore Logic

- **The Perpendicular Mandate:** All road bores shall be drawn at **90-degree angles**. Angled bores are strictly prohibited.
- **Pull Box Spacing:** Maximum footage between pull boxes is **650 feet**.
- **Drop Capacity:** Maximum of **6 drops** per conduit run. If exceeded, a second conduit is required.
- **Exclusion Zones:** Infrastructure shall **never** be placed in brick pavers or "Downtown Business Districts" where drops would require destroying concrete or blowing out the front of businesses.
- **Property Lines:** Handholes and flower pots shall be placed on property lines to minimize parcel obstruction.

Slack and Maintenance Loop Standards

- **Splitter HH:** 15-foot loop per splitter (Cyan).
- **HH Splice Locations:** 50-foot loop at each 3-way splice location (Cyan).
- **Aerial Coils:** 200-foot storage loops every 1,500–2,000 feet minimum.
- **Transitions:** 100-foot aerial coil at riser poles (aerial to buried).
- **Fiber Markers:** Placed every **500 feet** following the maintenance slack guideline, not at every handhole.

4. Fiber Architecture and Splicing Standards

The "Path of Light" philosophy dictates that "Fixed Position" logic is non-negotiable. Shifting counts to fill gaps is prohibited.

The Splitter Hierarchy and Naming

A single XGSPON port services 32 customers.

- **Naming Syntax:** [NODE]_[FEEDER]_[LEVEL][PORT]_[CITY CLLI][STATE] (e.g., OLT11_3P_7S_FTMFL).
- **IT Stack Requirements:** The 4-character City CLLI and 2-character State identifier are mandatory for database synchronization.
- **Ring Fiber Logic:** Named using the **12 IN / 12 OUT** convention (e.g., LE1_OLT07A: 1-12).

The "No Indexing" and "XD" Mandate

Fibers shall remain in the exact color position they occupied prior to the splice.

- **Rule:** Utilize the **highest non-split feeder counts first** (e.g., fiber 176 used before fiber 1).
- **XD Notation:** Utilized counts are replaced with "XD" dead count notations to maintain strand alignment.
- **Color-for-Color Exception:** If a new cable is smaller than the incoming count (e.g., placing fiber 49 into a 48-count cable), fiber 49 (a "blue" sequence fiber) shall be placed in the **Blue position (Position 1)** to maintain color alignment.

Feeder and Distribution Separation

- **80% Capacity Rule:** Upsize cable once 80% capacity is reached.
- **Separation:** Distribution fibers must come **after** non-split feeders in the tube.
- **Exception:** Front-loading distribution fibers is only permitted if utilizing the **very last non-split feeder fiber** in the cable.

Fiber Callouts

A. Introduction to Fiber Documentation Standards

Core Objectives The Lumos drafting standard is designed to achieve three non-negotiable goals:

- **Accuracy in Strand Alignment:** Every fiber must be accounted for and positioned precisely within its specific tube or ribbon.
- **Adherence to "Color for Color" Logic:** Maintaining a direct relationship between a fiber's assignment number and its physical color position to ensure field technicians can identify strands without guesswork.
- **Hierarchical Differentiation:** Establishing a clear visual and nomenclatural distinction between feeder fibers (transporting light from the cabinet) and distribution fibers (serving the individual address).

B. The Anatomy of a Lumos Fiber Callout

The structural formatting of a Lumos fiber callout is a technical requirement, not a stylistic preference. Technicians in the field rely on the absolute consistency of these annotations to perform complex splices under varied conditions.

Standard Naming Convention

Designers must identify every cable segment using the granular hierarchy established in our engineering deliverables. The following table deconstructs a standard cable string: HSP.01.03.14 48

Element	Description	Example
Market	Geographic market identifier.	HSP
Node	The specific OLT Node servicing the area.	01
Project	The assigned project number.	03
Segment	The specific cable segment ID.	14
Total Count	The physical capacity of the cable.	48

Punctuation and Mandatory Data Rules

- **Punctuation Requirements:** Designers **MUST** use commas to separate specific fiber blocks and semicolons to terminate cable-specific data strings (e.g., XD, 1-12;).
- **The "Same Line" Rule:** To ensure data integrity, the **Cable Name and the first associated fiber count** must reside on the same line.

- **The 20% Spare Mandate:** Every cable must represent at least **20% spare capacity** utilizing the "XD" count. This ensures future-proofing and limits the need for disruptive future overlays.

C. Hierarchical Naming and Splitter Identification

Hierarchical naming is the only way to maintain the "Path of Light" traceability from the Optical Line Terminal (OLT) to the customer's residence. This logic allows a maintenance technician to trace a failure at a specific address back through the secondary and primary splitters to the exact feeder fiber at the cabinet.

Deconstructing the Naming Logic

Consider the secondary splitter string: OLT11_3P_7S_FTMFL, 25-28

1. **Node (OLT11):** The specific field node or CO site.
2. **Feeder Fiber Assignment (3):** The specific feeder fiber (port) arriving from the cabinet.
3. **Splitter Type (P/S):** Identifies the hierarchy level (P = Primary 1x8, S = Secondary 1x4).
4. **Port Assignment (7):** The specific port off the primary splitter feeding this secondary unit.
5. **CLLI Code/State (FTMFL):** The 4-character city CLLI and 2-character state identifier.
6. **Output Range (25-28):** The distribution fibers serving the individual addresses.

D. The "No Indexing" Mandate and the "XD" Notation

Lumos strictly enforces a **"No Indexing"** policy. Fiber counts must never be moved up a tube or ribbon at a splice point to fill empty positions. Indexing leads to "wasteful splicing" and breaks the color-coding logic required for field operations. (note exception below)

The Role of "XD" Notation

"XD" notation is the placeholder for spare, reserved, or utilized counts. Its purpose is to maintain **fixed-position fiber indexing**. When a fiber is utilized at a splitter, it is replaced with "XD" in the outgoing callout to keep the remaining fibers in their original physical color positions.

Scenario Comparison: "Color for Color" in Practice

- **Incoming Callout:**

HSP.01.03.14 48

XD, 1-2;

OLT01_103P_FQVRNC, 3-48;

- **Outgoing Callout (Utilizing 3-7):**

HSP.01.03.14 48

XD, 1-7;

OLT01_103P_FQVRNC, 8-48;

The Color Sequence: To maintain "Color for Color" alignment, designers must adhere to the standard 12-color sequence: **1-Blue, 2-Orange, 3-Green, 4-Brown, 5-Slate, 6-White, 7-Red, 8-Black, 9-Yellow, 10-Violet, 11-Pink, 12-Aqua.**

E. Splicing Order of Operations and Technical Exceptions

Standardized sequencing is required for constructible prints. Lumos mandates that **Non-split Feeder fibers must be placed before Distribution (Split) fibers** in any callout.

The "Color for Color" Exception (Wrap-around Logic)

The "No Indexing" rule is only superseded when a new cable is physically too small to hold original assignment numbers. For example, if fiber 49 is being spliced into a 48-count cable, the physical position 49 does not exist. Because fiber 49 is inherently the first fiber of the next 48-count set (a "blue" fiber), it must be placed in the **first available blue position (Position 1)**. This wrap-around is triggered solely because the physical position cannot be utilized.

The Front-Loading Exception and Cable Sizing

Distribution fibers may only be **front-loaded** (placed at the start of a cable) if the **very last non-split feeder fiber** in that cable has been utilized.

F. Visual Implementation: Splice Boxes and Drafting Standards

Splice Point (SP) boxes are the visual instructions for construction crews. They must be drafted with absolute clarity.

"IN" vs. "OUT" Structure and Summation

- **IN:** Identifies incoming cable names, counts, and assignments.

- **OUT:** Identifies all outgoing cables and splitter tails.
- **Mandatory Calculation:** Designers **MUST** verify that the **sum of outgoing fibers accounts for all incoming strands**, including XD spares.

CAD Drafting Checklist

- **Orientation:** All callouts **MUST** be perfectly horizontal.
- **Background Masks:** Mandatory for all text to ensure legibility over map features.
- **Leaders:** Must snap to the closest corner of the splitter or HH block.
- **Case:** ALL CAPS for all text.
- **Indentation Rule:** If a data string for a single cable overflows to a second line, the "subset child line" must be indented to show it belongs to the cable listed above.
- **Footage Notation:** Use H-B (Handhole to Bend), B-F (Bore Pit to Flowerpot), or B-H (Bore Pit to Handhole). The "B" notation for a **Bore Pit** is critical for calculating labor costs in the Labor Span Sheet.

G. Quality Control: Identifying and Correcting Common Callout Errors

QC is the final safeguard against project failure. Because feeder counts propagate downstream, a single error in Sheet 1 can necessitate a total redesign of the entire job.

Common Errors and Solutions

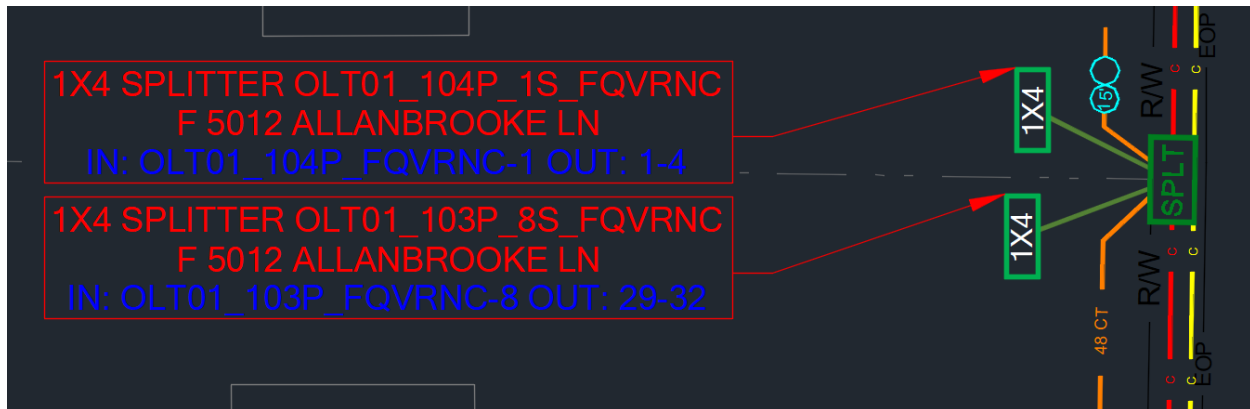
Error	Correction
"XD" used as "Available" (A-count)	The "A-count" was discontinued because it led to "unnecessary splicing." Use "XD" exclusively to maintain fixed positions.
Overlapping Text	Apply background masks; ensure no overlap with road names, dimensions, or parcel lines.
Misspelled Street Names	Manually entered names are frequent failures. Cross-check with EOP labels and Google Earth.
Indexing After Splitter	(Almost) Never move counts up. Replace utilized counts with "XD" and maintain original color position.
Incorrect Footage Abbreviation	Ensure "B" is used for any span involving a bore pit to ensure accurate labor billing.

The application of these standards ensures the long-term integrity of the Lumos fiber network. By following these prescriptive rules, we provide a professional, constructible deliverable that minimizes waste and maximizes network reliability.

Conduit Footage Abbreviations

Abbreviation	Meaning	Billing/Labor Impact
H-B / F-B	HH/FP to Bend	Parallel conduit; billed as one unit with main run.
B-F	Bend to FP	Bore shot ; isolated per-foot labor charge.
H-F / F-F	Origin to Destination	Total footage for combined bore and parallel segments.

5



. Quality Control and Submission Protocol

The professional appearance of a print is a proxy for technical accuracy. Designers shall perform a self-17-point QC review sequence before submission:

1. **Static Viewport Check:** Scroll all pages; no entities shall shift.
2. **Title Page:** Verify Lat/Long (second value must be negative), Legend, and Revision Table.
3. **Sheet Index:** Verify order and ensure **Wipeout frames are turned OFF**.
4. **Mini Map:** Consistency check across all sheets.
5. **Matchlines:** Accuracy, alignment, and centering on streets.
6. **Street Names:** Correct spelling, centering, and **no periods** in abbreviations.

7. **Linework:** Verify EOP corners are filleted and ROW is trimmed.
8. **Dimensions:** Parallelism and snap accuracy; no overlaps.
9. **Addresses:** Readability, background masks, and no overlap.
10. **Blocks:** All blocks brought to front and oriented correctly.
11. **Callouts:** Horizontal alignment and correct address/street spelling.
12. **Footages:** Correct abbreviations (H-B, B-F) and background masks.
13. **TO CO Arrows:** Required on every page containing FOC.
14. **North Arrow:** Required on every print sheet.
15. **VL/C Notations:** "VL" for vacant lots and "C" for commercial properties.
16. **Hatching:** Verify Gray (NN) or Yellow (Private) hatches; no hatching on public roads.
17. **Design Efficiency:** Verify maximum 6 drops per run and no T-connections in conduit.

The designer holds final responsibility for providing an efficient, professional, and constructible design. There is no excuse for non-compliance.